Strategic Technical Assessment and Remediation Roadmap for Ovulite: Leveraging Small-Data Architectures in Bovine IVF

1. Introduction and Project Mandate

The Ovulite initiative represents a significant ambition within the realm of precision livestock farming: the transition of bovine In Vitro Fertilization (IVF) from a subjective, artisanal practice dependent on individual technician skill into a quantifiable, data-driven science. The stated objective is to deploy a multi-modular ecosystem capable of AI-driven embryo grading, pregnancy success prediction, and optimal donor-recipient matching. This vision aligns with the broader industry trajectory toward "Farming 4.0," where biological intuition is augmented by computational intelligence.

However, the realization of such a system is inextricably linked to the underlying data substrate. The request at hand presents a specific, constrained scenario: the development of these advanced modules based on the "DayZee Farms" dataset—a collection of tabular records derived from real-world operations—in the absence of embryo video data and without the temporal runway to acquire new biological markers. This imposes a rigorous "cold start" problem often encountered in clinical AI deployment: how to build robust decision support systems (DSS) when the training data is sparse, multimodal inputs are missing, and ground truth is unbalanced.

This report serves as a definitive technical blueprint for this specific scenario. It moves beyond a superficial review to conduct a forensic audit of the provided data artifacts, evaluating their statistical power against the requirements of Deep Learning (DL) and Large Language Models (LLMs). It rigorously interrogates the feasibility of fine-tuning generative models for numerical prediction on small datasets and proposes a scientifically superior alternative: the utilization of Tabular Foundation Models (TFMs), specifically Prior-Data Fitted Networks (TabPFN), which are mathematically engineered for the low-data regime ($N < 1,000$) characterizing the current project status. Furthermore, it outlines a "Synthetic-to-Real" transfer learning strategy to address the absence of computer vision data, ensuring that the Ovulite architecture remains viable, scalable, and scientifically defensible despite current limitations.

2. Forensic Data Audit and Gap Analysis

To accurately calibrate the Ovulite system, we must first deconstruct the provided "DayZee

Farms" dataset. This audit evaluates the data not just for what is present, but for its latent statistical signal, consistency, and alignment with the proposed modules.

2.1 Structural Decomposition of the Data Ecosystem

The provided dataset comprises four primary CSV artifacts derived from the ET Summary.xlsx workbook. A line-by-line analysis reveals a highly fragmented but biologically specific record of operations.

2.1.1 The Core Training Corpus (ET Data.csv) The central repository for training any predictive model is the ET Data.csv file. A quantitative assessment reveals a dataset of significant sparsity. The file contains exactly **34 unique embryo transfer records**.[1] In the context of machine learning, where robust gradient-boosted trees typically require thousands of examples and deep neural networks require tens of thousands, a sample size of 34 is categorized as "ultra-small."

The outcome distribution within these 34 records is heavily unbalanced, a common characteristic of clinical datasets but a challenge for algorithmic training. Of the transfers recorded, only **9 resulted in confirmed pregnancies** (approximately 26.5%), while 25 were recorded as "Open" (failed).[1] This profound class imbalance implies that any model trained on this data without sophisticated resampling or prior-based regularization will likely collapse into predicting the majority class (failure), achieving a deceptively high accuracy ("73.5%") while having zero predictive power for the actual target of interest: success.

The feature set attached to these records provides a mixture of categorical and continuous variables. We observe Recipient ID, CL Size (measured in millimeters), Embryo Stage (e.g., Stage 7), Embryo Grade (e.g., Grade 1), Sire Name, Donor ID, and Protocol Type (e.g., "CIDR+E2+PGF+ECG").[1] Crucially, the dataset captures the variability of the "human factor" through the ET Tech column, identifying technicians (e.g., "Yaseen," "Hamid").[1] This is a valuable feature, as technician proficiency is a known non-biological variable affecting IVF success rates.[2]

2.1.2 The Recipient Context (Recipients Data.csv) While this file contains approximately 465 rows, a forensic examination of the indices reveals that the vast majority are tracking records for synchronization protocols rather than completed transfer events. For instance, rows corresponding to Protocol 1 (Indices 1-15) exhibit a systematic missingness in the Embryo Stage and Embryo Grade columns, denoted by a period ..[1] This effectively renders these rows unusable for training an embryo-centric prediction model, although they remain valuable for analyzing synchronization efficiency (e.g., response to CIDR) independent of embryo transfer.

Biological markers in this file are limited to CL Size (Corpus Luteum diameter) and CL Side. Explicit measurements for **Progesterone (P4)** and **Blood Flow** (Doppler vascularity indices), which were highlighted in the Ovulite presentation as critical "Sweet Spot" determinants, are

entirely absent from the CSV data.[1] This represents a significant deviation between the proposed system architecture and the available ground truth.

2.1.3 The Outcome Validation Sets (Client Reports) The files labeled for specific clients—Sardar Fahad, AQ Cattle, and Dawood—function as "positive-only" registries. For example, the AQ Cattle report lists 34 confirmed pregnancies [1], and the Dawood report lists 12.[1] While these records provide essential genetic traceability (linking Sires like "Mr. Clayton Slugger" to successful outcomes), they lack the corresponding "negative" examples (the failed transfers from the same batches). In binary classification, learning "what works" is mathematically impossible without also learning "what fails." Therefore, these files serve as validation sets for genetic potential rather than training sets for pregnancy probability.

2.2 The "Gold Gem" Rating: A Nuanced Evaluation

You requested a rating of this dataset on a scale of 1-10 regarding its utility for Ovulite. This rating cannot be a single number; it must be bifurcated based on the intended use case.

Rating as a Training Dataset for Deep Learning: 3.5 / 10

As a substrate for training the high-parameter models proposed in the presentation (CNNs for grading, Deep Learning for prediction), this dataset is poor. The volume (N=34) is orders of magnitude below the threshold for statistical significance in neural network training. The complete absence of image and video data renders the "Computer Vision" module unimplementable in its direct form. The lack of P4 and blood flow data strips the "biologically interpretable" models of their primary inputs.

Rating as a System Design & Validation Asset: 8.0 / 10

Conversely, as a seed dataset for designing the system's logic and workflow, the data is excellent. It possesses "Real-World Veracity." Unlike clean academic datasets, this data captures the messy reality of farm operations: inconsistent technician logging, protocol variations (Natural Heat vs. CIDR), and specific genetic combinations. It proves that the farm *is* collecting the metadata necessary to build a Decision Support System (DSS), even if the volume is currently low. It validates the schema required for the database (e.g., linking Sire EPDs to Recipient CL Size).

2.3 Gap Analysis Summary

The analysis identifies three critical gaps that the Ovulite technical strategy must bridge:

1. **The Visual Void:** The "AI Embryo Grading" module relies on Convolutional Neural Networks (CNNs) to analyze morphology.[1] The dataset contains *zero* pixel data.
2. **The Small-N Problem:** The "Pregnancy Prediction" module aims to use XGBoost or Deep Learning. Training these on 34 rows will lead to catastrophic overfitting.[3]
3. **The Biomarker Deficit:** The "Matching" module presumes access to P4 and blood flow

data. The dataset only offers CL size as a proxy.

3. Scientific Validity of Proposed Architectures

The user asks: *"Can we fine-tune an LLM for IVF success prediction... based on the dataset provided?"* This question touches on the frontier of current AI research, specifically the intersection of Large Language Models (LLMs) and tabular data. A rigorous analysis of current literature (2024-2025) provides a definitive answer.

3.1 The Fallacy of Fine-Tuning LLMs for Numerical Prediction on Small Data

Fine-tuning a foundational LLM (like Llama-3-70B or GPT-4) on a dataset of 34 tabular records is **scientifically inadvisable** and likely to yield performance inferior to simple logistic regression.

Theoretical Mismatch: LLMs are autoregressive models trained to minimize the perplexity of text tokens. Their "intelligence" lies in semantic reasoning and syntactic pattern matching. When applied to tabular prediction, the numerical data must be serialized into text (e.g., "The CL size is 18mm, the Grade is 1..."). Research demonstrates that LLMs struggle to capture the specific inductive biases required for tabular decision boundaries—specifically, the non-linear manifolds that decision trees (like XGBoost) handle so effectively.[3]

The Curse of Dimensionality: A model with 70 billion parameters has an immense capacity to memorize. When fine-tuned on only 34 examples, the model will essentially "memorize" the training set perfectly (Overfitting) but fail to generalize to new, unseen embryos. It creates a "hallucination engine" that generates plausible-sounding but mathematically groundless probabilities.

Benchmarking Evidence: Recent benchmarks comparing LLMs to traditional tabular methods reveal that while LLMs can perform reasonable zero-shot classification on *semantic* tasks (e.g., classifying text reviews), they are consistently outperformed by Gradient Boosted Decision Trees (GBDTs) like XGBoost and CatBoost on numerical tabular tasks, especially in low-data regimes.[3] The computational cost of fine-tuning an LLM for this task is disproportionate to the meager (or negative) accuracy gains.

3.2 The Superior Alternative: Tabular Foundation Models (TabPFN)

If fine-tuning an LLM is the wrong path, what is the optimal solution? The answer lies in a novel class of architectures emerging in 2024/2025: **Tabular Prior-Data Fitted Networks (TabPFN)**.

Architectural Fit: Unlike standard neural networks that start with random weights and require thousands of examples to "learn" patterns via gradient descent, TabPFN is a transformer

model that has been *pre-trained* on millions of synthetic datasets. It effectively has "meta-learned" the mathematical laws of Bayesian inference.[6]

The "In-Context" Advantage: TabPFN operates similarly to how an LLM handles a prompt. You feed it your 34 historical records as the "context," and it predicts the outcome for the 35th record in a single forward pass—no training, no backpropagation, and no hyperparameter tuning required.[9]

Performance on Small Data: Research explicitly highlights TabPFN's dominance in the "Small Data" regime (N < 1,000). On datasets of this size, it statistically outperforms tuned XGBoost, Random Forests, and Deep Neural Networks because it leverages its pre-trained "priors" to avoid overfitting, providing robust, well-calibrated probabilistic outputs even with tiny sample sizes.[10]

Conclusion: For the Ovulite Pregnancy Prediction module, **TabPFN is the scientifically optimal engine.** It turns the project's weakness (small data) into a scenario where the chosen technology excels.

4. Remediation Roadmap: The "Rescue" Architecture

Given the constraints—no video, minimal data, and no time for collection—we propose a pivoted technical roadmap. This approach preserves the functional requirements of the Ovulite modules while swapping the underlying implementation for methods that work with the available data.

4.1 Module 1: Pregnancy Prediction Engine (The "Brain")

Current Constraint: N=34 training samples. Missing P4/Blood Flow.

Proposed Architecture: Ensemble of TabPFN + Biological Priors.

Step 1: Feature Engineering & Imputation

The raw data must be transformed into biologically dense features to maximize the signal-to-noise ratio for the TabPFN model.

- **CL Logic:** The raw CL Size (mm) should be converted into a Luteal_Adequacy_Score. Based on literature, a CL < 15mm is a strong negative predictor for pregnancy maintenance due to insufficient progesterone capacity.[12] A continuous variable is good, but a categorical flag (Risk/Normal) helps the model identify non-linear thresholds.
- **Protocol Encoding:** The Protocol strings (e.g., "CIDR+E2+PGF+ECG") must be one-hot encoded or ordinal encoded based on "intensity" or "estrus control level."
- **Genetic Interaction:** Create a feature that combines Sire BW EPD (Birth Weight Expected Progeny Difference) and Recipient Breed. This captures biological incompatibilities (e.g., large calf genetics in smaller local breeds) which are known risk

factors for dystocia and early embryonic loss.[14]

Step 2: The Prediction Core

Deploy **TabPFN** (Python tabpfn library).

- **Input:** The 34 historical rows (Recipient features, Embryo metadata, Sire/Donor data).
- **Operation:** When a user enters a new potential transfer, the model uses the historical context to generate a probability distribution: *"Pregnancy Success Probability: 68% (Confidence Interval: 55-81%)"*.
- **Why this works:** TabPFN provides *uncertainty quantification*. With only 34 records, the model will rightly output wider confidence intervals, which is a critical safety feature for clinical decision support. A confident "Yes/No" from an overfitted XGBoost would be dangerous; a probabilistic range from TabPFN is honest and actionable.

4.2 Module 2: Embryo-Recipient Matching (The "Matchmaker")

Current Constraint: Matching requires predicting outcomes for *pairs* (Embryo X on Recipient Y). We lack sufficient historical pairs to train a recommender system.

Proposed Architecture: Heuristic Scoring Algorithm (Weighted Bipartite Matching).

Instead of a machine learning model, we implement a **Constructed Heuristic Algorithm**. This approach is standard in operations research when data is scarce but domain rules are well-defined.

Step 1: The Scoring Functions

We define mathematical scoring functions based on the "Gold Standard" literature found in the research scrape.

- **Recipient Score (S_R):**

$$S_R = W_1 \cdot f(CL) + W_2 \cdot f(Sync) + W_3 \cdot f(History)$$

  - *CL Factor ($f(CL)$):* Research indicates the Corpus Luteum is the primary engine of pregnancy maintenance. We apply a non-linear penalty: If CL < 15mm, Score = 0 (Do Not Transfer). If CL > 25mm (Potential Cyst), Score = 50. If CL 18-22mm (Ideal), Score = 100.[12]
  - *Synchronization Factor ($f(Sync)$):* Penalize recipients who did not show standing heat or required "forced" synchronization without natural signs, as they have lower conception rates.[16]
- **Embryo Score (S_E):**

$$S_E = W_A \cdot f(Grade) + W_B \cdot f(Stage) + W_C \cdot f(Genetics)$$

- *Grade Factor:* Grade 1 = 100, Grade 2 = 75, Grade 3 = 40. This weighting reflects the empirically observed drop in pregnancy rates (approx. 15-20%) per grade level.[17]
- *Stage Factor:* Prioritize Stages 7 (Expanded Blastocyst) and 6 (Blastocyst) over Stage 4 (Morula) for fresh transfers, as they indicate higher developmental competence.[19]

Step 2: The Matching Logic

Use a **Greedy Algorithm** or **Stable Marriage Algorithm** extension:

1. Rank all available Embryos by S_E.
2. Rank all prepared Recipients by S_R.
3. Pair the highest-scoring Embryo with the highest-scoring Recipient.
   - *Scientific Justification:* While "compensatory" mating (putting a poor embryo in a good cow) is sometimes practiced, biological data suggests that maximizing the "best-to-best" pairing yields the highest herd-level conception rate, which is the primary KPI for the farm.[19]

4.3 Module 3: AI Embryo Grading (The Vision "Ghost")

Current Constraint: No images/video. The "Computer Vision" module cannot be trained.

Proposed Architecture: Synthetic Training & Metadata Grading Proxy.

Solution A: The Metadata Proxy (Immediate Utility)

Rebrand the module to **"Morphokinetic Metadata Assessment"**. Even without images, the *metadata* reveals quality.

- **Logic Check:** If Embryo Stage = 4 (Morula) but Age = Day 8, the embryo is developmentally retarded. A technician might grade it "1" based on visual compactness, but the *data* says it is slow-growing and has low viability.
- **Implementation:** Build a logic engine that flags these metadata inconsistencies. This provides immediate "AI-driven" value by catching human errors in grading without needing computer vision.[21]

Solution B: Synthetic Data Strategy (The "Gold Gem" Pivot)

To fulfill the project's vision of image analysis without local data, utilize **Generative AI** for "Sim-to-Real" transfer learning.

1. **Sourcing:** Use public datasets like the **Gardner/IVF-Worldwide** dataset or those from *Gomez et al.* [22], which contain human or bovine embryo images.
2. **Generation:** Train a **StyleGAN** or **Latent Diffusion Model** (LDM) on these public images

to generate thousands of *synthetic* bovine embryo images representing different grades (1 vs. 3).[24]

3. **Training:** Train a lightweight CNN (e.g., ResNet-18 or MobileNet) on this synthetic dataset.
4. **Application:** This serves as a "Pre-Trained" module. When DayZee Farms eventually installs a camera, this model will already have learned basic morphological features (blastocoel cavity, inner cell mass edges) and can be fine-tuned on real data with very few shots. This demonstrates architectural foresight and solves the "cold start" problem.

5. The "Gold Standard" Gap and Future Data Strategy

The current dataset is a "Bronze" standard. To elevate Ovulite to a "Gold Standard" predictive system, specific variables must be integrated into the future data collection roadmap. These recommendations are derived from the intersection of the provided research snippets and global best practices in bovine theriogenology.

5.1 The Missing Biomarkers

1. **Progesterone (P4):** The most critical missing link.
   - *Why:* P4 concentration at the time of transfer is the single highest correlate of pregnancy maintenance.[25]
   - *Requirement:* Collect serum or milk P4 levels on Day 0 (Estrus) and Day 7 (Transfer). Values <2.5 ng/mL on Day 7 should trigger an automatic "No Transfer" recommendation.[12]
2. **Corpus Luteum Blood Flow:**
   - *Why:* CL Size alone is an imperfect proxy. A large CL can be cystic or functionally dead. Color Doppler ultrasonography measuring the percentage of vascular perfusion predicts pregnancy 15-20% better than size alone.[27]
   - *Requirement:* Log "Vascularity Score" (1-5) or "% Perfusion" alongside CL size.
3. **Genomic Breeding Values (GEBVs):**
   - *Why:* The dataset lists Sires (e.g., "Jackpot"). Integrating their official genomic scores for "Daughter Pregnancy Rate" (DPR) or "Sire Conception Rate" (SCR) adds a layer of predictive power that transcends the individual transfer event.[29]

5.2 The "Gold Standard" Data Checklist

For Ovulite to achieve SOTA (State-of-the-Art) performance, the database schema must evolve to include:

- [ ] **Recipient:** Parity, Body Condition Score (BCS), Days Post-Partum, History of Retained Placenta/Metritis.
- [ ] **Embryo:** Time of First Cleavage (morphokinetics via time-lapse), Freeze-Thaw interval, Media Batch ID.
- [ ] **Outcome:** Pregnancy confirmation at Day 30 (Ultrasound heartbeat) AND Day 60 (to

track Early Embryonic Death rates).

6. Conclusion and Strategic Recommendation

The "DayZee Farms" dataset, while sparse, is not a dead end; it is a **foundational seed**. It lacks the volume for "Big Data" approaches but is perfectly suited for "Smart Data" architectures like **TabPFN** and **Heuristic Algorithms**.

Final Strategic Recommendations:

1. **Abandon LLM Fine-Tuning for Prediction:** It is the wrong tool for numerical probability on N=34. Use **TabPFN** to achieve state-of-the-art results on the small dataset immediately.
2. **Utilize LLMs for Interface Only:** Use the LLM (Llama-3) as the *chatbot interface* that queries the TabPFN backend. The LLM handles the user's question ("Is Recipient 12 a good match?"), while TabPFN handles the math.
3. **Deploy Synthetic Vision:** Use public datasets and Generative AI to build the *pipeline* for the grading module, proving the capability exists even before the farm's hardware catches up.
4. **Build the Rule-Based Matcher:** Implement the "Stable Marriage" algorithm with biological weights (CL size, Grade) to deliver immediate value to the farm's decision-making process.

By adopting this "Rescue Architecture," Ovulite transforms from a project blocked by data scarcity into a cutting-edge demonstration of **Foundation Models applied to Precision Agriculture**, solving the exact problem (small data in high-value biology) that defines the modern industry.

7. Deep Dive: The Biological and Algorithmic "Why"

To fully justify the proposed pivot, we must explore the second and third-order implications of the data gaps and the chosen solutions. This section synthesizes the biological mechanisms of bovine reproduction with the mathematical theory of the recommended algorithms.

7.1 The Biological Imperative: Why CL Size is a "Dangerous" Proxy

The reliance on Corpus Luteum (CL) size as the primary biomarker in the dataset—in the absence of Progesterone (P4) and Blood Flow—presents a specific biological risk that the algorithm must account for.

The Mechanism of Failure: In bovine reproduction, the CL secretes progesterone, which is essential for endometrial receptivity and embryo survival. A common assumption is that "Bigger CL = More Progesterone." However, research indicates that this correlation is only moderate ($r \approx 0.4 - 0.6$).[28] A CL can be large (e.g., >25mm) but contain a fluid-filled

lacuna (cavity) or be cystic, rendering it functionally incompetent. Conversely, a smaller, compact CL (16-18mm) might be highly vascularized and produce optimal P4 levels.[15]

The Ripple Effect on Prediction:

If the Ovulite model trains *linearly* on CL size (assuming bigger is always better), it may erroneously recommend recipients with cystic CLs, leading to transfer failure.

- **Algorithmic Correction:** The TabPFN model or Heuristic Scorer must treat CL size non-linearly. We must engineer a "Sweet Spot" feature. Literature suggests that pregnancy rates plateau or even dip when CL diameters exceed 25-30mm (indicative of cysts) or fall below 13-15mm (luteal insufficiency).[13]
- **Implication:** The "Fall-back Rule Engine" proposed in Section 4.1 is not just a coding safety net; it is a biological necessity to filter out the "False Large" CLs that a simple linear regression would miss.

7.2 The "Small Data" Paradox: Why TabPFN Beats XGBoost

The recommendation to use TabPFN over the industry-standard XGBoost is grounded in the "Bias-Variance Tradeoff" inherent to small datasets (N=34).

The XGBoost Failure Mode (Variance): Gradient Boosted Decision Trees (GBDTs) like XGBoost work by iteratively correcting errors. On a dataset of 34 rows, XGBoost will rapidly find specific, noisy rules that perfectly classify the 9 pregnant cows (e.g., "If Technician = Hamid AND Day = Tuesday, then Pregnant"). This is **overfitting**. The model learns the *noise* of the specific dataset rather than the *signal* of bovine reproduction. When deployed on a new cow, these specific rules will fail, leading to poor generalization.[32]

The TabPFN Advantage (Prior Knowledge):

TabPFN fundamentally alters the learning paradigm. It is a transformer pre-trained to perform Bayesian inference on *any* tabular distribution. It doesn't "learn" from scratch on the 34 rows; it uses them to "orient" itself within its vast library of pre-learned mathematical functions.

- **Mechanism:** It treats the dataset as a *context window* (similar to how GPT-4 reads a prompt). It asks: "Given these 34 examples of inputs and outputs, and my knowledge of 100 million other datasets, what is the most mathematically probable outcome for the 35th input?"
- **Result:** This approach acts as a massive regularizer. It ignores the "Hamid + Tuesday" noise because that pattern doesn't align with the general mathematical structures it learned during pre-training. It focuses on robust signals (like CL Size vs. Outcome), providing a "smooth" probability curve that generalizes far better than a jagged decision tree on small data.[9]

7.3 Synthetic Data: The Bridge to Computer Vision

The user's constraint—"no video data... not enough time to collect it"—is a classic "Cold Start" problem in Computer Vision. The proposed solution of **Synthetic Data Generation** is not merely a workaround; it is a strategic asset.

Sim-to-Real Transfer Learning: By generating synthetic images of bovine embryos using StyleGAN or Diffusion models trained on public datasets (like those from Gardner et al. [22]), we create a "proxy" training set.

- **Benefit 1:** The CNN learns the fundamental "grammar" of an embryo image: the spherical shape of the Zona Pellucida, the texture of the Trophectoderm, the density of the Inner Cell Mass.[24]
- **Benefit 2:** When DayZee Farms eventually provides even a *small* set of real images (e.g., 50 images), the model can be "fine-tuned" on this real data. A model pre-trained on 10,000 synthetic embryos and fine-tuned on 50 real ones will vastly outperform a model trained on 50 real ones from scratch. This strategy "future-proofs" the Ovulite architecture.[35]

8. Summary of Deliverables & Next Steps

To execute this roadmap efficiently, the following technical deliverables are recommended for the immediate "Sprint":

1. **Data Ingestion Pipeline:** Write a Python script to ingest the 4 CSVs, clean the "Protocol 1" missing values (imputation), and generate the "Sweet Spot" features (CL categories).
2. **TabPFN Integration:** Implement tabpfn for the prediction endpoint. Wrap it in a simple API (FastAPI) that accepts Recipient/Embryo JSON and returns a JSON probability.
3. **Matching Logic Implementation:** Code the weighted bipartite matching algorithm in Python. Allow the user to adjust the weights (e.g., "Prioritize Genetics" vs. "Prioritize Pregnancy Chance") via a slider in the UI.
4. **Synthetic Vision Demo:** Generate a static set of "Grade 1" and "Grade 3" synthetic images to populate the UI, demonstrating the *user experience* of the grading module even if the backend is currently a simulation.

This report confirms that while the provided dataset is statistically lightweight, the *project* is viable. The application of **Tabular Foundation Models** and **Biological Heuristics** transforms the data constraints into architectural decisions that are scientifically sound, innovative, and immediately deployable.

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